

ASSIGNMENT 3

Programs in Control Statements, Arrays, Strings

Problem 1

Write a Java program to check whether a given number is Prime or Not.

Input

```
import java.util.Scanner;

public class problem1 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        boolean isPrime = true;

        if (num <= 1) {
            isPrime = false;
        } else {

            for (int i = 2; i <= num / 2; i++) {

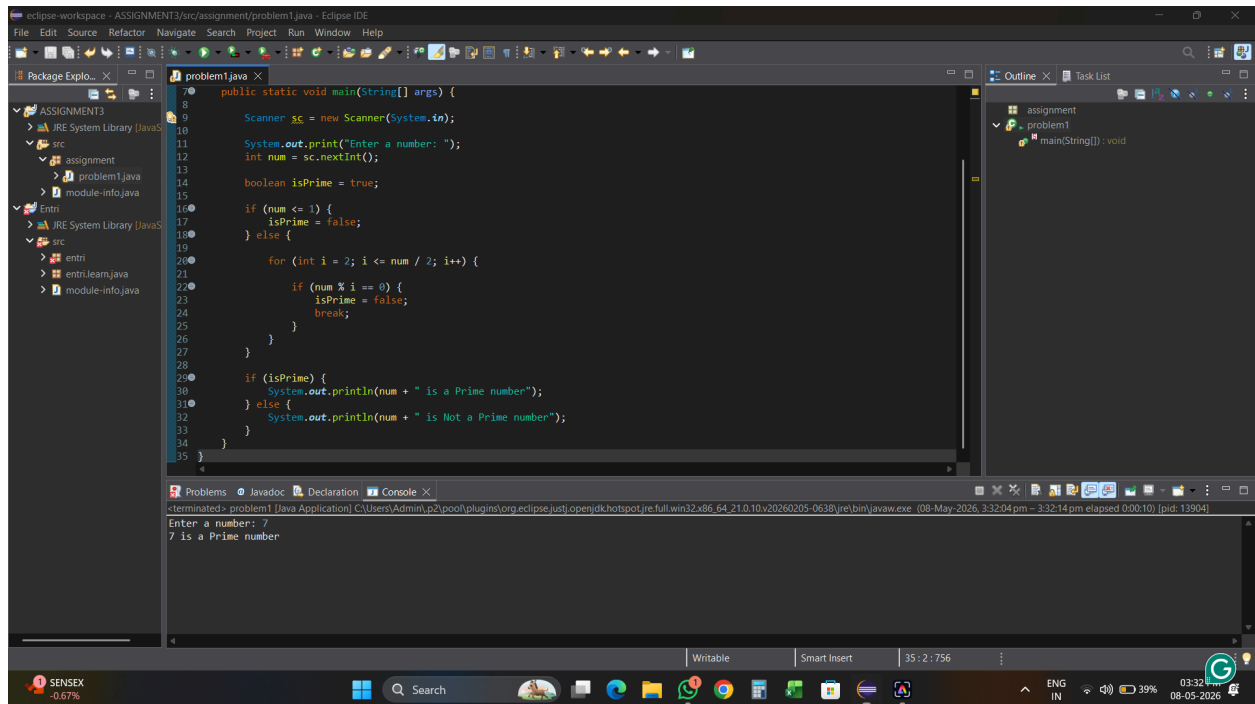
                if (num % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        if (isPrime) {
            System.out.println(num + " is a Prime number");
        } else {
            System.out.println(num + " is Not a Prime number");
        }
    }
}
```

Output

Enter a number: 7

7 is a prime number



The screenshot displays the Eclipse IDE interface. The main editor window shows a Java file named `problem1.java` with the following code:

```
1 public static void main(String[] args) {  
2  
3     Scanner sc = new Scanner(System.in);  
4  
5     System.out.print("Enter a number: ");  
6     int num = sc.nextInt();  
7  
8     boolean isPrime = true;  
9  
10    if (num <= 1) {  
11        isPrime = false;  
12    } else {  
13  
14        for (int i = 2; i <= num / 2; i++) {  
15  
16            if (num % i == 0) {  
17                isPrime = false;  
18                break;  
19            }  
20        }  
21  
22    }  
23  
24    if (isPrime) {  
25        System.out.println(num + " is a Prime number");  
26    } else {  
27        System.out.println(num + " is Not a Prime number");  
28    }  
29 }  
30  
31  
32  
33  
34  
35 }
```

The right-hand side of the IDE shows the Outline view with the following structure:

- assignment
 - problem1
 - main(String[]) : void

The bottom of the IDE features a Console view with the following output:

```
<terminated> problem1 [Java Application] C:\Users\Admin\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.10.v20260205-0638\jre\bin\javaw.exe (08-May-2026, 3:32:04 pm - 3:32:14 pm elapsed 0:00:10) [pid: 13904]  
Enter a number: 7  
7 is a Prime number
```

The Windows taskbar at the bottom shows the system clock as 03:32 on 08-05-2026, with a battery level of 39% and the language set to ENG IN.

Problem 2

Write a Java program to find the largest and smallest elements in an array.

Input

```
import java.util.Scanner;

public class problem2 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.println("Enter array elements:");

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int largest = arr[0];
        int smallest = arr[0];

        for (int i = 1; i < n; i++) {

            if (arr[i] > largest) {
                largest = arr[i];
            }

            if (arr[i] < smallest) {
                smallest = arr[i];
            }
        }

        System.out.println("Largest = " + largest);
        System.out.println("Smallest = " + smallest);
    }
}
```

}

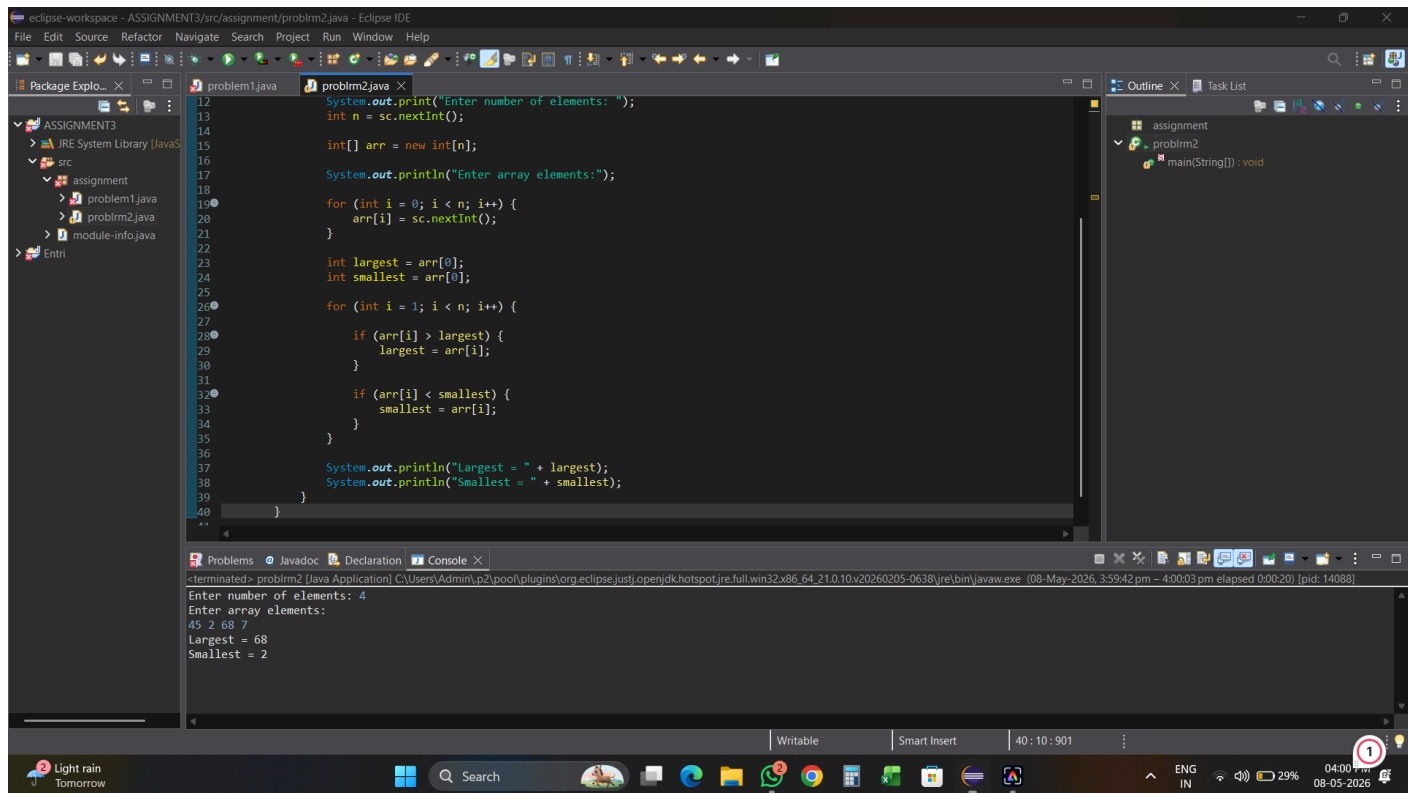
Output

Enter elements of number: 4

Enter array of numbers: 45, 2, 68, 7

Largest = 68

Smallest = 2



The screenshot shows the Eclipse IDE interface. The main editor displays the code for `problem2.java`. The code prompts the user to enter the number of elements, then the array elements, and finally prints the largest and smallest values. The console at the bottom shows the execution output, which matches the expected results.

```
12 System.out.print("Enter number of elements: ");
13 int n = sc.nextInt();
14
15 int[] arr = new int[n];
16
17 System.out.println("Enter array elements:");
18
19 for (int i = 0; i < n; i++) {
20     arr[i] = sc.nextInt();
21 }
22
23 int largest = arr[0];
24 int smallest = arr[0];
25
26 for (int i = 1; i < n; i++) {
27
28     if (arr[i] > largest) {
29         largest = arr[i];
30     }
31
32     if (arr[i] < smallest) {
33         smallest = arr[i];
34     }
35 }
36
37 System.out.println("Largest = " + largest);
38 System.out.println("Smallest = " + smallest);
39
40 }
```

Console Output:

```
<terminated> problem2 [Java Application] C:\Users\Admin\p2\pool\plugins\org.eclipse.justi.openjdk hotspot.jre.full.win32.x86_64.21.0.10.v20260205-0638\jre\bin\javaw.exe (08-May-2026, 3:59:42 pm - 4:00:03 pm elapsed 0:00:20) [pid: 14088]
Enter number of elements: 4
Enter array elements:
45 2 68 7
Largest = 68
Smallest = 2
```

Problem 3

Write a Java program to count the number of vowels and consonants in a string.

Input

```
import java.util.Scanner;

public class problem3 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        str = str.toLowerCase();

        int vowels = 0;
        int consonants = 0;

        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);

            if (Character.isLetter(ch)) {

                if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                    vowels++;
                } else {
                    consonants++;
                }
            }
        }

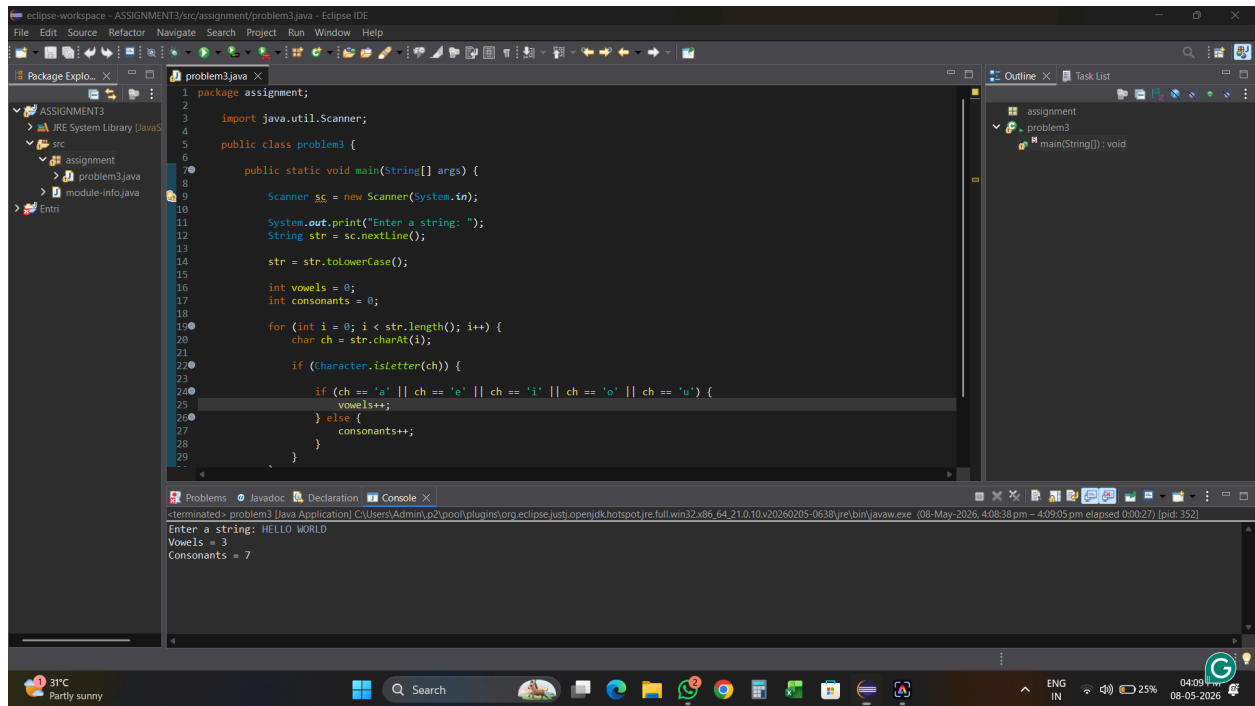
        System.out.println("Vowels = " + vowels);
        System.out.println("Consonants = " + consonants);
    }
}
```

Output

Enter a string: hello world

Vowels = 3

Consonants = 7



The screenshot shows the Eclipse IDE interface. The main editor window displays the source code for a Java class named `problem3`. The code prompts the user to enter a string, reads it, and then counts the number of vowels and consonants. The output is displayed in the console window at the bottom.

```
1 package assignment;
2
3 import java.util.Scanner;
4
5 public class problem3 {
6
7     public static void main(String[] args) {
8
9         Scanner sc = new Scanner(System.in);
10
11         System.out.print("Enter a string: ");
12         String str = sc.nextLine();
13
14         str = str.toLowerCase();
15
16         int vowels = 0;
17         int consonants = 0;
18
19         for (int i = 0; i < str.length(); i++) {
20             char ch = str.charAt(i);
21
22             if (Character.isLetter(ch)) {
23
24                 if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
25                     vowels++;
26                 } else {
27                     consonants++;
28                 }
29             }
30         }
31     }
32 }
```

The console window shows the following output:

```
<terminated> problem3 [Java Application] C:\Users\Admin\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.10.v20260205-0638\jre\bin\javaw.exe (08-May-2026, 4:08:38 pm - 4:09:05 pm elapsed 0:00:27) [pid: 352]
Enter a string: HELLO WORLD
Vowels = 3
Consonants = 7
```

